



The value-creating potential of AI: A multi-dimensional analysis of effects and mechanisms

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ABSTRACT

The integration of artificial intelligence in corporate operations has emerged as a transformative force in business landscapes, yet existing research provides limited insights into the mechanisms through which AI creates enterprise value. Drawing on a comprehensive dataset of Chinese listed companies spanning 2010–2022, this study investigates how AI adoption influences enterprise value through the mediating role of data assets and the moderating effect of managerial capability. Our analysis reveals that AI adoption significantly enhances enterprise value both directly and indirectly through data assets, demonstrating substantial mediation effects. The findings establish that managerial capability positively moderates this relationship, indicating that firms with stronger management teams extract greater value from AI investments. Through extensive robustness checks and endogeneity tests, we establish the causal nature of these relationships. Three heterogeneity tests reveal boundary conditions: the valuation premium is stronger for state-owned enterprises, disappears in heavily polluting industries unless AI is part of a credible green transition, and is evident only in non-asset-intensive firms where resource reconfiguration is agile. By integrating resource-based, dynamic-capability, and stakeholder perspectives, the study clarifies when AI becomes a genuine source of competitive advantage. The results inform executives that data governance, leadership development, and asset or environmental restructuring must accompany AI budgets, and they guide policymakers toward incentive designs that couple digital adoption with sustainability and capital-efficiency objectives.

1. Introduction

The digital revolution is no longer a distant horizon but an immediate operational reality reshaping how firms generate, appropriate, and sustain value. Artificial intelligence (AI) now occupies the epicentre of this upheaval, serving simultaneously as an innovation enabler, a process optimiser, and a strategic decision partner across virtually every sector, from heavy manufacturing and logistics to financial services and creative industries (Bloom, Prettnner, Saadaoui, & Veruete, 2025; Bonaparte, 2024; Xin, 2025; Yang, 2025). Global spending on AI-centric solutions eclipsed USD 150 billion in 2024 and is forecast to double within four years, signalling that boardrooms increasingly view machine learning, natural-language processing, and computer vision not as peripheral IT utilities but as core capabilities essential for long-term competitiveness (Durongkadej, Hu, & Wang, 2024; Kurup & Gupta,

2022; Zhou, Li, Wang, Li, & Zhou, 2025). Yet this widespread enthusiasm is tempered by persistent execution gaps: surveys reveal that only a minority of deployments move beyond experimental pilots into scaled, revenue-impacting applications, while failure rates for enterprise AI projects still exceed 50 % (Han, Li, & Jin, 2025; Lee, Kim, & Ivan, 2023). Early adopters such as leading e-commerce and fintech platforms demonstrate that data-driven algorithms can compress cycle times, personalise customer journeys, and unlock entirely new revenue streams, but many traditional incumbents struggle to translate proof-of-concept successes into firm-wide performance gains (Yin, Yin, & Wen, 2025). These mixed outcomes suggest that the value-creating capacity of AI depends on more than technological sophistication alone; it hinges on complementary intangible resources, effective governance structures, and context-specific managerial capabilities—factors that current empirical research has only partially explored.

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The evolution of AI applications has transcended traditional operational boundaries, catalyzing unprecedented changes in organizational structures and business models (Dong, Xin, Liu, & Yu, 2024). As Singh (2021) observes, AI's capacity for superhuman performance in specific domains has revolutionized sectors ranging from healthcare to manufacturing. This technological advancement has given rise to novel business architectures, particularly in platform-based models and service-oriented frameworks, necessitating a deeper examination of their value-creation mechanisms (Lee, Suh, Roy, & Baucus, 2019).

A critical yet underexplored aspect of AI adoption lies in its intersection with organizational resources and capabilities. Our research posits that the relationship between AI adoption and enterprise value is inherently mediated by an organization's data assets and moderated by managerial capabilities (Dai, 2025a; Li, Zhong, & Tong, 2024; Wamba-Taguimdje, Wamba, Kamdjoug, & Wanko, 2020). This perspective addresses a significant gap in current understanding, as highlighted by recent studies (Enholm, Papagiannidis, Mikalef, & Krogstie, 2022; Perifanis & Kitsios, 2023).

The synthesis of AI capabilities with organizational resources presents unique challenges and opportunities. Evidence suggests that organizations excelling in data resource utilization while fostering innovation-oriented leadership demonstrate superior outcomes in AI adoption (Enholm et al., 2022; Wamba-Taguimdje et al., 2020). The emergence of generative AI technologies has further amplified the importance of sophisticated decision-making frameworks, underlining the critical role of management expertise in technology deployment (Jiang, Xuan, & Zhang, 2024).

To empirically examine these relationships, we employ a methodologically rigorous approach utilizing a comprehensive dataset of Chinese listed companies spanning 2010–2022. Our study introduces an innovative measurement framework that captures the multifaceted nature of organizational AI adoption. The primary measurement leverages natural language processing techniques, specifically a Word2vec-based dictionary methodology, to quantify AI adoption levels from corporate disclosures. This approach processes corporate annual reports through a Skip-gram model, identifying and weighting 73 AI-related terms based on semantic relevance. To ensure measurement robustness, we complement this with a patent-based approach that tracks firms' AI-related intellectual property development through the Chinese Patent Office database. This dual measurement strategy enables us to capture both the strategic emphasis and technological advancement dimensions of corporate AI initiatives (Li et al., 2024). Drawing from the CNRDS Platform, we construct comprehensive measures for our mediating variable (data assets) and moderating variable (managerial capability), alongside a robust set of firm-level controls to account for potential confounding effects.

This research advances the theoretical discourse on AI-driven value creation through several key contributions. First, we illuminate the specific pathways through which AI adoption influences enterprise value. Second, we demonstrate how data assets serve as crucial intermediaries in translating AI investments into tangible outcomes (Lui, Lee, & Ngai, 2022; Naqvi, 2021). Third, we establish the pivotal role of managerial capability in optimizing AI deployment effectiveness (Abou-Foul, Ruiz-Alba, & López-Tenorio, 2023).

Our findings yield significant implications for practitioners and policymakers alike. For corporate leaders, we provide evidence-based guidance for optimizing AI investments through enhanced data management and leadership development. For policymakers, our results inform the development of frameworks that foster innovation while addressing the ethical considerations inherent in AI adoption (Zhang, Song, & Wang, 2023). The complexities of AI integration in organizational contexts demand a nuanced understanding of its impact on performance and value creation. As enterprises navigate this technological frontier, our research provides crucial insights for developing sustainable competitive advantages while addressing the broader implications of AI adoption in the business landscape (Jiao, Lu, Ren, & Wei, 2024).

2. Literature review and hypothesis development

2.1. AI and enterprise value creation

The integration of artificial intelligence (AI) into business operations has emerged as a transformative force, fundamentally altering how organizations create and capture value. AI technologies enable firms to automate processes, enhance decision-making, and optimize resource allocation, thereby improving operational efficiency and effectiveness (Cao, 2021). The potential of AI to drive enterprise value creation is particularly evident in its ability to analyze vast amounts of data, uncover insights, and facilitate informed strategic decisions (Nurlia, Daud, & Rosadi, 2023). For instance, AI-driven analytics can identify market trends and consumer preferences, allowing organizations to tailor their offerings and enhance customer experiences (Dai, 2025b; Fischer, 2024).

Moreover, the co-creation of value through AI is increasingly recognized as a critical component of successful business strategies. By leveraging AI technologies, firms can engage customers in the value creation process, fostering collaboration and enhancing customer satisfaction (Kaartemo & Helkkula, 2018). This shift toward a more interactive and participatory approach to value creation aligns with the principles of service-dominant logic, which emphasizes the importance of relationships and interactions in generating value (Swan, Peltier, & Dahl, 2024). As organizations adopt AI, they not only improve their internal processes but also create opportunities for external stakeholders to participate in value co-creation, thereby enhancing overall enterprise performance (Baabdullah, 2024).

The relationship between AI adoption and enterprise value is further supported by empirical evidence demonstrating that organizations that effectively utilize AI technologies experience significant improvements in performance metrics such as revenue growth, market share, and customer loyalty (Ghaffar, Arshad, Siddique, & Nasir, 2024). However, the realization of these benefits is contingent upon the organization's ability to integrate AI into its existing business models and processes, highlighting the need for strategic alignment and organizational readiness (Alami et al., 2021).

In summary, AI serves as a catalyst for enterprise value creation by enhancing operational efficiency, enabling data-driven decision-making, and fostering collaborative value co-creation with customers and stakeholders. As organizations continue to invest in AI technologies, understanding the mechanisms through which AI contributes to value creation becomes increasingly important for both practitioners and researchers.

2.2. The mediating role of data assets

Data assets play a pivotal role in the effective implementation of AI technologies and their subsequent impact on enterprise value (Eichler, Gröger, Hoos, Schwarz, & Mitschang, 2022; Jian, Xiaofan, Jinlin, & Yunqing, 2023). The ability to collect, manage, and analyze data is essential for organizations seeking to leverage AI for competitive advantage (Kashyap, Morse, Patel, & Shah, 2021). Data serves as the foundation upon which AI algorithms operate, and the quality and quantity of data directly influence the effectiveness of AI applications (Nurlia et al., 2023). Organizations that possess robust data assets are better positioned to extract actionable insights from AI systems, leading to improved decision-making and enhanced operational performance.

Furthermore, the integration of data assets with AI technologies facilitates the development of predictive models and advanced analytics, enabling organizations to anticipate market trends and customer behaviors (Sun, Xu, Yu, & Wang, 2022). This predictive capability not only enhances strategic planning but also allows firms to respond proactively to changes in the business environment, thereby creating additional value (Cao, 2021). For example, companies that utilize AI-driven analytics to optimize supply chain management can reduce costs, improve

inventory turnover, and enhance customer service levels (Ghaffar et al., 2024).

The mediating role of data assets in the AI-value creation relationship is supported by research indicating that organizations with strong data management capabilities are more likely to realize the full potential of their AI investments (Nurlia et al., 2023). By effectively leveraging data, firms can enhance their AI applications, leading to improved performance outcomes and increased enterprise value. Therefore, understanding the interplay between AI adoption and data assets is crucial for organizations aiming to optimize their value creation strategies.

2.3. Managerial capability as a boundary condition

Managerial capability is a critical boundary condition that influences the effectiveness of AI adoption and its impact on enterprise value (Helfat & Martin, 2015). The successful adoption of AI technologies requires not only technical expertise but also strong leadership and strategic vision (Huang, 2024). Managers play a vital role in guiding the organization through the complexities of AI integration, ensuring that AI initiatives align with overall business objectives and are supported by appropriate resources (Kashyap et al., 2021).

Research has shown that organizations with capable management teams are better equipped to navigate the challenges associated with AI adoption, including issues related to data governance, ethical considerations, and change management (Kaartemo & Helkkula, 2018). Effective managerial practices can facilitate a culture of innovation and adaptability, enabling organizations to respond to the dynamic nature of AI technologies and market demands (Swan et al., 2024). Moreover, managers who prioritize employee training and development in AI-related skills can enhance workforce readiness and engagement, further contributing to the successful implementation of AI initiatives (Nurlia et al., 2023).

The moderating effect of managerial capability on the AI-value creation relationship is underscored by studies indicating that organizations with strong leadership and strategic alignment are more likely to achieve positive outcomes from their AI investments (Weber, Engert, Schaffer, Weking, & Krcmar, 2023). By fostering an environment that encourages collaboration, experimentation, and continuous learning, managers can enhance the effectiveness of AI technologies and maximize their potential for value creation.

2.4. Hypothesis development

Based on the theoretical foundations established in the preceding sections, we propose the following hypotheses to guide our empirical investigation:

Hypothesis 1. *AI adoption demonstrates a positive association with enterprise value.*

We hypothesize that the implementation of AI technologies has a direct positive effect on enterprise value. Organizations that effectively integrate AI into their operations are expected to experience improvements in performance metrics such as revenue growth, operational efficiency, and customer satisfaction.

Hypothesis 2. *Data assets mediate the relationship between AI adoption and enterprise value.*

We propose that data assets mediate the relationship between AI adoption and enterprise value. Specifically, organizations with strong data management capabilities will be better positioned to leverage AI technologies, leading to enhanced decision-making and improved performance outcomes.

Hypothesis 3. *Managerial capability moderates the relationship between AI adoption and enterprise value.*

We hypothesize that managerial capability moderates the

relationship between AI adoption and enterprise value. Organizations with capable management teams are expected to achieve greater benefits from AI investments, as effective leadership can facilitate the successful integration of AI technologies and promote a culture of innovation. In summary, this theoretical review highlights the intricate relationships between AI adoption, data assets, managerial capability, and enterprise value creation. By investigating these dynamics, our research aims to contribute to a deeper understanding of how organizations can optimize their AI strategies to enhance value creation.

3. Research design

3.1. Data and sample selection

Our empirical investigation examines Chinese listed companies from 2010 to 2022, comprising 22,938 firm-year observations. This extensive timeframe captures both the emergence and maturation of AI adoption across Chinese enterprises. We retrieve data from the Chinese Research Data Services (CNRDS) Platform, which provides standardized financial and operational metrics. To ensure data quality, we exclude financial institutions due to their distinct regulatory environment, ST and *ST companies, firms with missing data for key variables, and observations with extreme values at the 1st and 99th percentiles.

3.2. Variable measurement

3.2.1. Primary independent variable: AI adoption

We develop a sophisticated measurement approach using Word2vec technology to quantify organizational AI adoption levels. The process begins with constructing an AI-specific dictionary through Skip-gram model training on corporate annual reports and patent texts. Through cosine similarity analysis, we identify and select terms most semantically related to our seed AI keywords. After eliminating duplicates, irrelevant terms, and low-frequency words, our final dictionary comprises 73 AI-related terms. The AI adoption score is calculated as:

$$AI_{Score} = \ln(1 + \sum AI_{term frequency})$$

For robustness, we construct an alternative measure based on AI-related patent applications. We identify patents containing keywords such as “artificial intelligence,” “intelligent,” and “automation” from the Chinese Patent Office database, calculating:

$$AI_{PatentScore} = \ln(1 + \sum AI_{patents})$$

3.2.2. Dependent variable: Enterprise value

The measurement of enterprise value through Tobin's Q represents a critical component of our research design, employing two distinct calculation methods to ensure comprehensive value capture. Our primary measure constructs Market Value A by combining A-shares market value, B-shares market value (converted to RMB), non-tradable shares value, and total liabilities. This approach effectively addresses the complexities of China's split-share structure while incorporating both equity and debt components of firm value. The inclusion of non-tradable shares, valued using the equity-to-registered capital ratio, reflects the unique ownership characteristics of Chinese listed companies and provides a more complete picture of corporate worth.

In parallel, we employ an alternative calculation focused on market-traded shares, where Market Value B emphasizes actively traded equity components while maintaining consistency in liability treatment. This complementary approach reduces potential upward bias that might arise from non-tradable share valuation, offering a more conservative estimate aligned with international valuation practices. The conversion of B-shares to RMB using appropriate exchange rates ensures currency consistency across both measurements.

The dual measurement strategy strengthens our analysis by

addressing potential biases inherent in single valuation methods while acknowledging the institutional characteristics of China's market environment. This methodological choice enhances result reliability through measurement triangulation, particularly important when examining the relationship between AI adoption and enterprise value. Both calculations normalize market value by total assets, maintaining the theoretical foundation of Tobin's Q as a ratio of market value to replacement cost, thereby facilitating meaningful cross-sectional comparison among firms of varying sizes.

3.2.3. Mediating variable: Data assets

We measure data assets through the proportion of data-related keywords in annual reports, capturing organizations' emphasis on data management and utilization. This measure reflects both the quantity and strategic importance of data resources within the organization.

3.2.4. Moderating variable: Managerial capability

Drawing from established literature, we construct a comprehensive measure of managerial capability incorporating management team characteristics, educational background, industry experience, and strategic decision-making effectiveness.

3.2.4.1. Control variables. Our empirical analysis incorporates a comprehensive set of control variables to account for firm-specific characteristics that might influence enterprise value. Firm size, measured as the natural logarithm of total assets, controls for scale effects and operational complexity. Debt-Asset ratio (total liabilities divided by total assets) captures financial structure and risk exposure, while revenue growth rate reflects operational momentum and market opportunities. Financial leverage, calculated as the ratio of long-term debt to equity, accounts for capital structure decisions and financial risk. Board size measures corporate governance effectiveness through the number of directors, complemented by ownership concentration (shareholding percentage of top three shareholders) to reflect governance structure and monitoring intensity. The SA index serves as a proxy for financial constraints, capturing firms' ability to access external financing. These controls collectively account for fundamental firm characteristics, governance mechanisms, and financial conditions that prior literature identifies as significant determinants of enterprise value, thereby isolating the unique effects of AI adoption on firm performance.

3.3. Econometric specifications

Our investigation into the relationship between AI adoption and enterprise value employs a systematic econometric approach. The baseline model examines the direct effect of AI adoption on enterprise value through the following specification:

$$Value_{it} = \beta_0 + \beta_{1AI_{it}} + \beta_{2Size_{it}} + \beta_{3DAR_{it}} + \beta_{4Growth_{it}} + \beta_{5FL_{it}} + \beta_{6Board_{it}} + \beta_{7Top3_{it}} + \beta_{8SA_{it}} + Year + Industry + \varepsilon_{it} \quad (1)$$

To examine the mediating role of data assets, we follow Baron and Kenny's approach with a three-step analysis. First, we test the relationship between AI adoption and data assets:

$$Data_{it} = \alpha_0 + \alpha_{1AI_{it}} + Controls_{it} + Year + Industry + \varepsilon_{it} \quad (2)$$

Second, we confirm the direct effect of AI adoption on enterprise value:

$$Value_{it} = \gamma_0 + \gamma_{1AI_{it}} + Controls_{it} + Year + Industry + \varepsilon_{it} \quad (3)$$

Third, we incorporate both AI adoption and data assets:

$$Value_{it} = \delta_0 + \delta_{1AI_{it}} + \delta_{2Data_{it}} + Controls_{it} + Year + Industry + \varepsilon_{it} \quad (4)$$

To investigate the moderating effect of managerial capability, we extend our model to include interaction terms:

$$Value_{it} = \theta_0 + \theta_{1AI_{it}} + \theta_{2MC_{it}} + \theta_{3AI_{it}} \times MC_{it} + Controls_{it} + Year + Industry + \varepsilon_{it} \quad (5)$$

In these specifications, i denotes individual firms and t represents time periods. We include both year and industry fixed effects to control for temporal and sector-specific variations. This comprehensive modeling approach enables systematic examination of direct effects, mediation pathways, and moderation impacts while controlling for firm-specific characteristics.

3.4. Summary statistics and multicollinearity test

The descriptive statistics presented in Table 1 reveal significant variations in AI adoption patterns across Chinese listed companies. The mean AI adoption score of 0.795 with a standard deviation of 1.125 suggests considerable heterogeneity in firms' technological advancement strategies. Enterprise value, measured by Tobin's Q, displays a mean of 1.900, indicating that sample firms generally trade above their book values, though with substantial variation ($SD = 0.981$). The data asset ratio maintains a relatively low mean of 0.005, highlighting that formal data asset recognition remains at an early stage among Chinese firms, consistent with Wang, Shi, Nevo, Li, and Chen (2015)'s findings regarding the emerging nature of data asset management practices.

The diagnostic results from Table 2 demonstrate the statistical validity of our model specification. The variance inflation factors (VIF) for all variables fall well below the conventional threshold of 5, with a mean VIF of 1.280. The highest individual VIF of 1.580 for AI adoption suggests minimal concern regarding multicollinearity. These results provide confidence in the independence of our predictor variables, allowing for reliable coefficient estimation. The relatively low VIF values for data assets (1.560) and managerial capability (1.060) indicate that these key variables capture distinct organizational characteristics, supporting our theoretical framework's construct validity.

4. Results and discussions

4.1. Baseline regression result

The baseline regression results presented in Table 3 offer compelling evidence about the relationship between AI adoption and enterprise value in Chinese listed companies, while revealing important insights about the control variables' effects.

The analysis demonstrates a consistently positive and statistically significant relationship between AI adoption and firm value across all three model specifications. In Model 1, which includes year fixed effects but not industry fixed effects, AI adoption shows a positive association with firm value. This relationship remains robust when incorporating industry fixed effects in Model 2 and when including both year and

Table 1
Descriptive statistics.

Variable	N	Mean	p50	SD	Min	Max
FV	22,938	1.900	1.604	0.981	0.837	7.826
AI	22,938	0.795	0	1.125	0	4.564
DATA	22,938	0.00500	0	0.0140	0	0.113
MA	22,938	2.461	2	1.113	1	4
Size	22,938	22.31	22.09	1.294	20.11	26.11
DAR	22,938	0.418	0.414	0.196	0.0580	0.849
Growth	22,938	0.158	0.108	0.331	-0.481	1.795
FL	22,938	1.248	1.051	0.791	-0.382	5.784
Board	22,938	2.126	2.197	0.195	1.609	2.639
lnTop3	22,938	3.823	3.867	0.343	2.858	4.443
SA	22,938	-3.817	-3.824	0.253	-4.396	-3.211

Table 2

VIF test.

Variable	VIF	1/VIF
AI	1.580	0.634
DAR	1.570	0.635
DATA	1.560	0.642
Size	1.540	0.649
lnTop3	1.130	0.886
FL	1.120	0.895
Board	1.090	0.920
SA	1.080	0.927
MA	1.060	0.944
Growth	1.030	0.967
Mean	1.280	

Table 3

Baseline regression.

VARIABLES	FV	FV	FV
AI	0.094*** (18.055)	0.019*** (2.771)	0.026*** (3.937)
Size	-0.240*** (-44.545)	-0.214*** (-35.923)	-0.229*** (-40.438)
DAR	-0.708*** (-20.169)	-0.468*** (-11.992)	-0.445*** (-12.208)
Growth	0.233*** (13.776)	0.247*** (14.222)	0.215*** (13.086)
FL	-0.046*** (-6.247)	-0.038*** (-5.005)	-0.041*** (-5.797)
Board	0.009 (0.297)	-0.021 (-0.679)	0.012 (0.424)
lnTop3	-0.078*** (-4.665)	-0.047*** (-2.605)	-0.014 (-0.870)
SA	0.202*** (8.124)	0.195*** (7.929)	0.201*** (8.072)
Constant	8.841*** (53.731)	8.198*** (39.818)	8.261*** (25.201)
Observations	22,938	22,938	22,938
R-squared	0.294	0.260	0.361
IND	NO	FE	FE
YEAR	FE	NO	FE

Note: t-statistics in parentheses; *** p < 0.01, ** p < 0.05, * p < 0.1.

industry fixed effects in Model 3. The persistence of this positive relationship across different specifications aligns with [Luo and Yu \(2022\)](#) findings regarding the value-enhancing effects of AI technologies in emerging markets, supporting our first hypothesis about AI adoption's positive impact on enterprise value.

The control variables reveal consistent patterns across all model specifications. Firm size demonstrates a significant negative relationship with enterprise value, suggesting that smaller firms in our sample generally achieve higher market valuations relative to their assets. The debt-asset ratio (DAR) shows a strong negative association with firm value, indicating that higher leverage levels may constrain firm value creation, consistent with [Su's \(2010\)](#) observations about capital structure effects in Chinese firms. Growth opportunities, measured by revenue growth rate, exhibit a robust positive relationship with firm value, confirming the market's positive valuation of growth prospects.

The model's explanatory power improves substantially when incorporating both industry and year fixed effects, with the R-squared increasing from 0.294 in Model 1 to 0.361 in Model 3. This improvement suggests that industry-specific characteristics and temporal factors play crucial roles in determining firm value, emphasizing the importance of controlling for these effects. Financial leverage (FL) maintains a consistent negative relationship across all specifications, while board size shows no significant impact, suggesting that this governance mechanism might not be a crucial determinant of firm value in our sample. The SA index, measuring financial constraints, displays a positive relationship with firm value, potentially indicating that firms with certain financial constraints might be more efficient in their resource

allocation.

These findings extend beyond mere statistical significance to provide meaningful economic insights. The consistent positive relationship between AI adoption and firm value, even after controlling for various firm characteristics and fixed effects, suggests that markets recognize and value firms' AI initiatives. This aligns with [Wang, Liao, and Wang \(2024\)](#)'s argument that technological advancement serves as a crucial driver of firm value in contemporary markets. The results also underscore the complex interplay between financial structure, growth opportunities, and firm value in the context of technological advancement.

4.2. Addressing endogeneity issues

The endogeneity test in [Table 4](#) employs a two-stage least squares (2SLS) regression to establish causality between AI adoption and enterprise value. The highly significant relationship between lagged and current AI adoption in the first stage ($t = 135.526$) validates our choice of instrumental variable. This strong persistence suggests that AI adoption decisions reflect structured, long-term strategic planning rather than opportunistic technological investments.

The second-stage results retain statistical significance while controlling for potential reverse causality, strengthening the causal interpretation of AI's impact on firm value. The instrumental variable approach reveals that after accounting for endogeneity concerns, the positive relationship between AI adoption and firm value remains robust ($t = 2.894$). The reliability of these findings is supported by the model's substantial explanatory power, with R-squared values of 0.782 and 0.376 for the first and second stages respectively.

Interestingly, the control variables display distinct patterns across the two stages. While firm size positively influences AI adoption in the first stage ($t = 5.907$), it maintains a negative relationship with firm value in the second stage ($t = -36.604$). This contrasting effect suggests that larger firms are more likely to adopt AI technologies, yet size itself may not translate directly into higher market valuations. Similarly, growth opportunities show positive associations in both stages but with varying magnitudes, indicating that high-growth firms may leverage AI adoption more effectively for value creation.

Additionally, our findings are further validated through two sophisticated econometric techniques in [Table 5](#): Propensity Score Matching

Table 4

Endogeneity check: instrumental variable approach.

	(1)	(2)
VARIABLES	AI	FV
L.AI	0.788*** (135.526)	
AI		0.026*** (2.894)
Size	0.025*** (5.907)	-0.240*** (-36.604)
DAR	-0.023 (-0.873)	-0.467*** (-11.676)
Growth	0.064*** (4.481)	0.230*** (11.617)
FL	-0.001 (-0.222)	-0.046*** (-8.488)
Board	-0.004 (-0.184)	0.018 (0.574)
lnTop3	0.001 (0.067)	0.066*** (3.842)
SA	-0.008 (-0.408)	0.285*** (10.617)
Constant	-0.433* (-1.912)	8.557*** (35.813)
Observations	18,648	18,648
R-squared	0.782	0.376
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 5
Endogeneity check: PSM and Heckman approach.

VARIABLES	PSM	Heckman
	FV	FV
AI	0.027*** (3.088)	0.130*** (11.224)
imr1		1.102*** (10.879)
Size	−0.230*** (−28.744)	−0.524*** (−18.904)
DAR	−0.468*** (−9.172)	−1.056*** (−15.778)
Growth	0.242*** (10.579)	0.485*** (16.304)
FL	−0.049*** (−4.698)	−0.113*** (−11.637)
Board	0.056 (1.369)	0.013 (0.448)
lnTop3	−0.035 (−1.491)	−0.200*** (−8.398)
SA	0.199*** (5.677)	0.191*** (7.688)
Constant	8.082*** (19.538)	14.794*** (21.636)
Observations	12,230	22,938
R-squared	0.370	0.364
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(PSM) and Heckman's two-stage procedure. The PSM analysis, based on 12,230 matched observations, confirms the positive relationship between AI adoption and firm value ($t = 3.088$). This matching approach effectively controls for selection bias by comparing firms with similar characteristics but different AI adoption levels. (See Table 6.)

The Heckman correction method, utilizing the full sample of 22,938 observations, yields stronger statistical evidence of AI adoption's impact on firm value ($t = 11.224$). The significant inverse Mills ratio (imr1, $t = 10.879$) indicates the presence of selection effects and justifies the use of this correction technique. Both approaches maintain comparable R-squared values (0.370 and 0.364 respectively), suggesting consistent

Table 6
Robustness checks.

VARIABLES	(1)	(2)	(3)	(4)
	FV	FV_replace	FV	FV
AI_replace	0.001*** (3.351)			
AI		0.040*** (4.136)	0.025*** (3.216)	0.047*** (5.801)
Size	−0.230*** (−40.261)	−0.385*** (−45.993)	−0.349*** (−26.953)	−0.254*** (−37.644)
DAR	−0.442*** (−12.129)	−1.068*** (−19.812)	0.258*** (5.109)	−0.407*** (−9.407)
Growth	0.216*** (13.182)	0.558*** (22.989)	0.157*** (11.218)	0.093*** (4.878)
FL	−0.041*** (−5.794)	−0.052*** (−4.869)	−0.026*** (−4.167)	−0.041*** (−5.066)
Board	0.009 (0.298)	−0.086** (−1.980)	0.040 (0.916)	−0.024 (−0.704)
lnTop3	−0.015 (−0.928)	0.433*** (17.597)	−0.315*** (−10.206)	−0.015 (−0.774)
SA	0.200*** (8.030)	0.693*** (18.797)	0.880*** (10.048)	0.169*** (5.670)
Constant	8.294*** (25.269)	12.427*** (25.612)	14.014*** (30.326)	8.806*** (25.837)
Observations	22,915	22,938	22,938	15,623
R-squared	0.361	0.419	0.266	0.418
IND	FE	FE	NO	FE
YEAR	FE	FE	FE	FE
ID	NO	NO	FE	NO

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

model explanatory power across different specifications.

The endogeneity analysis thus confirms that the value-enhancing effects of AI adoption stem from fundamental organizational transformation rather than spurious correlation or reverse causality. These results provide strong empirical support for inferring a causal relationship between AI adoption and enterprise value.

4.3. Robustness checks

The robustness tests employ four alternative approaches to validate our main findings. Model 1 uses an alternative measure of AI adoption based on patent counts (AI_replace), maintaining statistical significance ($t = 3.351$) while demonstrating a positive relationship with firm value. This alternative measurement approach confirms that our findings are not driven by the specific construction of our primary AI adoption metric.

Model 2 employs an alternative calculation of enterprise value (FV_replace), revealing a stronger positive association with AI adoption ($t = 4.136$). The increased R-squared (0.419) suggests that this alternative value measure might capture additional aspects of firm performance. Control variables show notably different magnitudes in this specification, particularly for ownership concentration (lnTop3) which shifts to a strong positive association ($t = 17.597$).

Model 3 introduces firm fixed effects while excluding industry fixed effects, demonstrating that the positive relationship between AI adoption and firm value persists ($t = 3.216$). Interestingly, the debt-asset ratio (DAR) shows a positive relationship in this specification, contrasting with its negative effect in other models, suggesting that within-firm variation in leverage might have different implications than cross-sectional differences.

Model 4 excludes COVID-19 period observations, focusing on the pre-pandemic sample (15,623 observations). The stronger coefficient significance for AI adoption ($t = 5.801$) indicates that the relationship might have been partially masked by pandemic-related market disruptions. The model's higher R-squared (0.418) suggests better explanatory power for the pre-pandemic period.

These varied specifications collectively reinforce the robustness of our main findings, demonstrating that the positive relationship between AI adoption and enterprise value remains stable across different variable measurements, model specifications, and sample periods.

4.4. The mediating effect of DATA

The mediation result in Table 7 provides compelling evidence supporting both H1 and H2 through a systematic examination of the relationships between AI adoption, data assets, and firm value. Following Baron and Kenny's framework, our analysis reveals several key findings. The results show that AI adoption significantly influences data assets ($t = 66.512$), indicating that firms with greater AI adoption tend to develop more substantial data resources. This relationship explains nearly half of the variation in firms' data asset levels, as evidenced by the R-squared value of 0.489.

When examining the joint impact of AI adoption and data assets on firm value, both factors demonstrate significant positive effects ($t = 1.997$ and $t = 3.969$ respectively). The reduced magnitude of AI adoption's coefficient in the presence of data assets, while maintaining statistical significance, indicates partial mediation. This finding suggests that data assets serve as an important channel through which AI adoption enhances enterprise value, while AI adoption simultaneously maintains a direct positive effect on firm value.

Control variables exhibit consistent patterns throughout the analysis, with firm size, leverage, and financial constraints showing significant relationships with firm value. The stability of these effects across model specifications reinforces the reliability of the mediation results. Notably, ownership concentration and financial constraints display negative associations with data assets, suggesting that firms with concentrated

Table 7
Mediating effect of DATA.

	(1)	(2)
VARIABLES	DATA	FV
AI	0.006*** (66.512)	0.014** (1.997)
DATA		2.045*** (3.969)
Size	0.000 (1.490)	−0.229*** (−40.488)
DAR	0.000 (0.168)	−0.445*** (−12.217)
Growth	−0.000 (−0.338)	0.215*** (13.099)
FL	−0.000 (−0.476)	−0.041*** (−5.786)
Board	0.000 (0.866)	0.012 (0.402)
lnTop3	−0.001*** (−5.149)	−0.012 (−0.734)
SA	−0.001*** (−3.310)	0.203*** (8.160)
Constant	−0.005 (−1.230)	8.272*** (25.240)
Observations	22,938	22,938
R-squared	0.489	0.3614
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

ownership or financial limitations might invest less in data resources. These comprehensive findings demonstrate that while AI adoption directly contributes to firm value, it also operates indirectly by fostering the development and utilization of valuable data assets, thus illuminating a crucial mechanism in the value creation process.

The bootstrap test results in Table 8, based on 500 replications, provide rigorous statistical validation of both direct and indirect effects in our mediation analysis. The direct effect of AI adoption on firm value shows strong statistical significance ($Z = 10.26$, $p < 0.001$) with a coefficient of 0.0483. The 95 % confidence interval [0.0392, 0.0578] excludes zero, confirming the robustness of AI adoption’s direct impact on enterprise value.

The indirect effect through data assets demonstrates similar statistical significance ($Z = 6.38$, $p < 0.001$) with a coefficient of 0.0405. Its confidence interval [0.0287, 0.0528] also excludes zero, confirming the significance of the mediation pathway. The comparable magnitudes of direct (0.0483) and indirect (0.0405) effects suggest that data assets mediate approximately 45.6 % of AI adoption’s total effect on firm value, highlighting the substantial role of data assets in translating AI capabilities into enterprise value. These bootstrap results strengthen our previous findings by demonstrating their robustness to sampling variation and providing precise estimates of the mediation effects. The results quantitatively validate both the direct value-creating effect of AI adoption and its indirect influence through enhanced data asset utilization, supporting our theoretical framework regarding the dual pathways through which AI adoption enhances enterprise value.

4.5. The moderating effect of ECON

In Table 9, the analysis examines the moderating role of managerial

Table 8
Bootstrap test.

	Coefficient	Standard error	Z value	P value	95 % confidence interval	
Direct effect	0.0483291	0.0047117	10.26	0.000	0.039205	0.0577913
Indirect effect	0.0404704	0.0063463	6.38	0.000	0.0286665	0.0528445
Replication:500						

Table 9
Moderating effect of MA.

	(1)	(2)
VARIABLES	FV	FV
AI	0.026*** (3.965)	0.027*** (4.127)
MA x AI		0.012** (2.517)
MA	0.051*** (8.742)	0.053*** (8.930)
Size	−0.223*** (−39.241)	−0.223*** (−39.209)
DAR	−0.452*** (−12.420)	−0.453*** (−12.464)
Growth	0.185*** (11.025)	0.184*** (10.998)
FL	−0.039*** (−5.529)	−0.040*** (−5.549)
Board	0.016 (0.563)	0.017 (0.567)
lnTop3	−0.033** (−1.979)	−0.035** (−2.077)
SA	0.219*** (8.797)	0.218*** (8.747)
Constant	8.138*** (24.844)	8.140*** (24.851)
Observations	22,938	22,938
R-squared	0.363	0.3633
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

capability through two model specifications. Model 1 establishes that both AI adoption and managerial capability exhibit significant positive effects on firm value when examined separately. Model 2 extends this analysis by incorporating an interaction term (MA x AI), which shows statistical significance while maintaining the significance of both individual variables. This positive interaction effect provides strong support for our third hypothesis, indicating that managerial capability enhances the relationship between AI adoption and firm value.

The control variables demonstrate consistent effects across both specifications. In both models, firm size and debt-asset ratio maintain significant negative relationships with firm value, while growth and the SA index show positive associations. Financial leverage consistently displays negative effects, and ownership concentration shows moderate negative significance. Board size remains insignificant across both specifications.

The models incorporate both industry and year fixed effects, with similar explanatory power as indicated by their R-squared values. This comprehensive analysis, based on 22,938 observations, demonstrates that firms with stronger managerial capability are better positioned to translate AI investments into enterprise value, suggesting that management quality plays a crucial role in realizing the potential benefits of AI technologies.

4.6. Heterogeneity analysis

4.6.1. Ownership heterogeneity

Splitting the sample by property rights reveals that artificial-intelligence investment lifts firm value in both state-owned enterprises (SOEs) and non-SOEs, but with substantially different elasticities. As

Table 10
Ownership Heterogeneity.

VARIABLES	Non-SOE	SOE
	FV	FV
AI	0.019** (2.327)	0.035*** (3.160)
Size	−0.221*** (−26.011)	−0.267*** (−32.330)
DAR	−0.403*** (−8.066)	−0.639*** (−11.932)
Growth	0.251*** (11.658)	0.131*** (5.190)
FL	−0.047*** (−4.283)	−0.031*** (−3.488)
Board	−0.066 (−1.636)	0.069 (1.588)
lnTop3	−0.040* (−1.769)	0.050* (1.854)
SA	0.100*** (2.822)	0.458*** (12.068)
Constant	8.171*** (21.012)	9.583*** (30.884)
Observations	14,079	8381
R-squared	0.320	0.466
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

shown in Table 10, for SOEs the AI coefficient is 0.035 ($t = 3.160$, $p < 0.01$), almost double the 0.019 recorded for non-SOEs ($t = 2.327$, $p < 0.05$). This suggests that, once Chinese capital markets price in size, leverage, and other fundamentals, state ownership magnifies the shareholder gains attributable to digital technologies. Two complementary mechanisms are plausible. First, SOEs typically command easier access to subsidised financing and policy support, allowing them to scale data-intensive projects quickly and to absorb short-term implementation frictions. Second, AI deployments in state firms often align with government imperatives—such as industrial upgrading and public-service efficiency—so stock-market participants may anticipate both an earnings boost and lower political risk. (See Table 11.)

The control variables reinforce this interpretation. The debt-to-asset ratio exerts a stronger negative drag on SOEs (−0.639) than on non-SOEs (−0.403), implying that investors penalise inefficient leverage

Table 11
Pollution-intensity heterogeneity.

VARIABLES	Non-heavily Polluting	Heavily Polluting
	FV	FV
AI	0.027*** (3.732)	0.023 (1.307)
Size	−0.219*** (−32.481)	−0.263*** (−26.840)
DAR	−0.481*** (−11.172)	−0.301*** (−4.657)
Growth	0.202*** (10.530)	0.238*** (7.703)
FL	−0.039*** (−4.208)	−0.044*** (−4.302)
Board	−0.031 (−0.881)	0.152*** (2.949)
lnTop3	0.006 (0.310)	−0.092*** (−2.883)
SA	0.180*** (6.189)	0.308*** (6.695)
Constant	8.019*** (19.898)	9.645*** (26.341)
Observations	17,799	5139
R-squared	0.351	0.377
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

more heavily when the state is the ultimate guarantor; yet the positive AI effect offsets part of that discount. Growth opportunities carry a larger premium in non-SOEs (0.251 versus 0.131), consistent with private firms' greater agility, while the significant SA coefficient for SOEs (0.458) highlights how internal cash constraints matter less when external funding is readily available. Ownership concentration changes sign across groups: higher blockholder control depresses non-SOE valuations (−0.040) but adds value in SOEs (0.050), perhaps because concentrated stakes curb agency slippage in bureaucratic hierarchies.

Finally, the explanatory power of the model jumps from an R-squared of 0.320 in the non-SOE panel to 0.466 among SOEs, indicating that the chosen covariates—and especially the AI variable—capture more of the variation in market performance for firms under state control. Together, these patterns corroborate the view that institutional context conditions the payoff to advanced analytics: AI brings measurable benefits across ownership types, yet state backing amplifies the transmission of those benefits to equity markets.

4.6.2. Pollution-intensity heterogeneity

Partitioning the sample by pollution intensity shows that stock markets reward artificial-intelligence spending only when environmental liabilities are modest. In non-heavily polluting firms the AI coefficient is 0.027 ($t = 3.732$, $p < 0.01$), indicating that every marginal rise in the AI-adoption index translates into a clear premium in Tobin's Q after controlling for size, leverage, growth, and governance characteristics. By contrast, the coefficient falls to 0.023 and loses statistical significance in heavily polluting enterprises ($t = 1.307$). Investors thus appear reluctant to capitalise digital-transformation stories when future cash flows are clouded by stricter emission standards, possible carbon-pricing liabilities, and higher capex needs for end-of-pipe technologies.

The pattern of the controls supports this reading. Leverage exerts a softer discount among polluters (−0.301 versus −0.481), hinting that creditors already price in regulatory backing or implicit guarantees, whereas equity holders remain wary of residual risk. Board size is positive and significant only in the high-pollution group (0.152), suggesting that a larger director pool adds monitoring value when firms face complex compliance landscapes. Ownership concentration turns negative (−0.092) for these firms, consistent with the idea that minority investors demand greater voice to safeguard against environmental contingencies.

Growth prospects carry a slightly larger premium among polluters (0.238 versus 0.202), implying that the market welcomes signals of future revenue expansion that might offset remediation costs. Still, the muted AI coefficient implies that digital technologies alone do not reassure investors unless accompanied by credible environmental transition strategies.

The R-squared climbs marginally from 0.351 to 0.377 when moving from clean to polluting industries, showing that our covariates explain more of the valuation variance where regulatory exposure is pronounced. Overall, the evidence suggests that environmental context conditions the financial return to AI: in cleaner sectors digital tools are seen as productivity enhancers, whereas in pollution-intensive settings their value story is overshadowed by sustainability risk.

4.6.3. Asset-intensity heterogeneity

The evidence in Table 12 shows that artificial-intelligence spending lifts firm value only when a company's business model is non-asset-intensive; once heavy fixed capital dominates the balance sheet, the market no longer prices AI as a clear source of superior cash flow. Resource-based reasoning explains this asymmetry: light-asset firms depend on intangibles that can be redeployed quickly, so predictive analytics and automation can reshape cost structures and product offerings without long payback periods. By contrast, asset-intensive enterprises are locked into lengthy depreciation cycles and rigid production systems, slowing the sensing-and-reconfiguring routines through which digital investments normally enhance competitiveness.

Table 12
Asset-intensity Heterogeneity.

VARIABLES	Non-asset-intensity industry	Asset-intensity industry
	FV	FV
AI	0.030*** (4.228)	−0.009 (−0.492)
Size	−0.230*** (−35.500)	−0.225*** (−21.116)
DAR	−0.431*** (−10.406)	−0.495*** (−7.020)
Growth	0.215*** (11.555)	0.187*** (5.766)
FL	−0.041*** (−4.733)	−0.040*** (−3.494)
Board	−0.015 (−0.451)	0.127** (2.235)
InTop3	−0.005 (−0.272)	−0.069** (−2.028)
SA	0.182*** (6.458)	0.304*** (6.125)
Constant	8.248*** (22.797)	8.775*** (21.176)
Observations	19,023	3915
R-squared	0.351	0.383
IND	FE	FE
YEAR	FE	FE

Note: t-statistics in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Investors therefore treat AI in these settings as an incremental tool rather than a game-changing capability.

For managers operating in capital-heavy industries, the message is that standalone AI projects will not shift market expectations unless they dovetail with broader moves—such as modularising equipment, adopting digital-twin architectures for plant optimisation, or pivoting toward asset-light service models—that unlock the organization from physical rigidity. Only when digital programmes are visibly integrated with strategies that shorten asset-turnover cycles and release trapped capital will shareholders expect tangible value creation.

5. Conclusion

5.1. Key research findings

This study employed a 2010–2022 panel of 22,938 firm-year observations from Chinese listed companies to untangle how artificial-intelligence initiatives influence corporate market value. Two complementary measurement strategies—a Word2vec-based disclosure index and counts of AI-related patent grants—captured both the strategic salience and the technological depth of each firm's digital commitment. After controlling for firm size, leverage, governance, industry, and year effects, AI adoption displayed a statistically and economically meaningful association with higher Tobin's Q. Almost one-half of this valuation premium is channelled through the build-up and exploitation of proprietary data assets, underscoring that algorithmic capabilities alone are insufficient: only when accompanied by data stewardship do they translate into credible earnings prospects.

Managerial capability strengthens this process. Firms whose leadership teams possess deep analytical expertise and strategic agility convert AI expenditure into shareholder value more efficiently than peers, corroborating the dynamic-capabilities proposition that sensing, seizing, and reconfiguring routines are human-centred. Heterogeneity analyses sharpen the insight. First, state-owned enterprises (SOEs) capture a larger AI-related valuation gain than non-SOEs, suggesting that institutional sponsorship and preferential financing lower the perceived risk of digital transformation. Second, AI elevates firm value in non-heavily polluting sectors but not in high-pollution industries, indicating that environmental liabilities can offset the expected efficiency gains unless digital projects are explicitly tied to sustainability

outcomes. Third, the technology boosts market value only in non-asset-intensive firms; in asset-heavy contexts, long depreciation cycles and rigid capital structures slow the reconfiguration of resources, muting the impact of predictive analytics and automation. Collectively, these findings extend resource-based, stakeholder, and institutional perspectives by showing that digital capability yields competitive advantage only when either structural flexibility is high or state backing cushions adjustment costs.

5.2. Managerial and policy implications

For corporate executives and boards, AI expenditures should be embedded in an integrated change programme that also fortifies data governance, upgrades leadership skills, and—where physical or environmental lock-ins are severe—redesigns core assets. In light-asset, low-pollution businesses, rolling out machine-learning systems can be a stand-alone efficiency lever. By contrast, capital-intensive or emission-prone firms need to pair AI with modularisation of equipment, digital-twin architectures for plant optimisation, or circular-economy product redesign to convince investors that digital tools will affect long-run cash flows. Boards should demand evidence that AI projects advance enterprise-wide dynamic capabilities rather than remain confined to isolated pilot teams.

For investors, The results highlight the value of a conditional screening approach. The same AI disclosure can imply markedly different cash-flow paths depending on a firm's ownership structure, pollution exposure, and capital rigidity. Portfolio managers may refine ESG and digital-transformation ratings by weighting AI indicators more heavily in flexible or state-supported settings while discounting them in asset-locked or high-emission contexts unless coupled with credible transition plans.

For policymakers, A one-size-fits-all subsidy for AI adoption risks misallocation. Light-asset sectors may respond adequately to generic R&D tax credits or procurement preferences, whereas capital-heavy or pollution-intensive industries require blended instruments—such as concessional finance for green retrofits, performance-based carbon allowances, and data-infrastructure grants—that jointly lower environmental and digital barriers. Enhanced disclosure guidelines for AI-enabled sustainability metrics would also help capital markets differentiate between superficial automation spending and genuine capability upgrades.

5.3. Study limitations

Several caveats temper the generalisability of our conclusions. First, the institutional environment of China—characterised by an active industrial policy, bank-dominated finance, and frequent state shareholding—may amplify the valuation impact of AI relative to other jurisdictions. Second, although the study deployed instrumental-variable, propensity-score-matching, and Heckman corrections, unobserved factors such as organizational culture or unreported pilot failures could still confound the results. Third, the AI index blends text analytics and patent counts; it may overstate dormant patents or under-capture operational machine-learning systems that firms choose not to patent. Fourth, the analysis relies on market-based value metrics; future operating-performance gains or losses remain to be tracked. Finally, annual data obscure the timing of investment waves and the gestation lags typical of large-scale digital programmes.

5.4. Directions for future research

Cross-country comparative studies could reveal how legal origin, capital-market maturity, and data-governance regimes moderate the AI–value linkage. Fine-grained, project-level expenditure logs combined with milestone completion data would allow researchers to separate immediate announcement effects from longer-run cash-flow realization.

Longitudinal case studies in asset-heavy, pollution-intensive firms could unpack micro-processes by which digital twins, robotics, and predictive maintenance interact with plant modernisation and decarbonisation journeys. Future work might also explore nonlinear or threshold effects—whether a critical mass of AI spending must be reached before markets reassess a firm's growth trajectory, or whether excess investment triggers diminishing returns. Finally, integrating labour-force microdata would illuminate how AI-driven productivity interacts with wage structures, skill upgrading, and internal career mobility, offering a more holistic view of value creation that spans capital and human resources.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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